ANGLE GRINDER BLADES (METAL CUTTING & POLISHING)

1. Description

Angle grinder blades (also called discs) are attachments used with an angle grinder for cutting, grinding, and polishing metal surfaces.

- **Cutting Discs**: Thin, abrasive discs designed for cutting metal rods, pipes, and sheets.
- **Grinding/Polishing Discs**:
  - Grinding wheels: Used for removing excess material and smoothing rough edges.
  - Flap discs / polishing pads: Used for finishing, polishing, and surface preparation.

Common Materials:

- Aluminum oxide
- Silicon carbide
- Zirconia alumina

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2. How to Operate

Setup

1. Disconnect power before installing the blade.
2. Select the correct disc:
   - Thin disc → cutting
   - Thick/flap disc → grinding/polishing
3. Mount the disc securely using the flange nut.
4. Ensure the guard is properly positioned.

Operation

1. Hold the grinder firmly with both hands.
2. Start the grinder and allow it to reach full speed.
3. Apply the blade gently to the workpiece:
   - Cutting: Use straight motion, no twisting.
   - Grinding: Use slight angle (~15–30°).
4. Do not apply excessive pressure—let the tool do the work.
5. Switch off and wait for the disc to stop before placing it down.

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3. Do’s and Don’ts

Do’s

- Use the correct disc for the job.
- Inspect discs for cracks before use.
- Maintain a steady grip.
- Work in a well-ventilated area.
- Use proper RPM-rated discs compatible with the grinder.

Don’ts

- Do not use damaged or worn-out discs.
- Do not remove the safety guard.
- Do not force the tool during cutting.
- Do not use cutting discs for grinding.
- Do not touch the disc immediately after use (it gets very hot).

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4. Safety Measures

- Wear **PPE**:
  - Safety goggles / face shield
  - Gloves
  - Ear protection
  - Safety shoes
- Keep sparks away from flammable materials.
- Ensure proper clamping of the workpiece.
- Maintain safe body positioning (avoid line of disc rotation).
- Disconnect power before changing discs.
- Use spark shields if working indoors.

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5. Maintenance

- Clean dust and metal particles after use.
- Store discs in a dry place to prevent moisture damage.
- Regularly check:
  - Spindle lock
  - Guard condition
  - Power cable
- Replace worn or uneven discs immediately.
- Lubricate moving parts of the grinder if required.

6. Real Images